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RESEARCH ARTICLE



From brain to behavior: Psychological resilience mediates associations between whole-brain resting-state connectivity and NSSI

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ABSTRACT

Background: Non suicidal self injury (NSSI) is a public health concern, and its prevalence has increased significantly following the COVID-19 pandemic. Despite its rising incidence, the neurobiological mechanisms underlying NSSI behaviour in adolescents remain poorly understood.

Methods: A sample of 89 adolescents (46 NSSI positive, 43 NSSI negative) aged 15.39 ± 1.77 years was recruited from clinical settings. NSSI behaviour and psychological resilience were evaluated. Resting-state functional magnetic resonance imaging (Rs-fMRI) was conducted to examine brain connectivity patterns. Data analysis incorporated descriptive and inferential statistics, as well as support vector machine algorithms, to identify the neural correlates of NSSI and resilience.

Results: The NSSI positive group had significantly lower resilience scores ($M = 23.41$, $SD = 7.95$). Connectivity between the sensorimotor and limbic networks was negatively associated with NSSI ($r = -0.222$, $p < 0.05$), while connectivity between the sensorimotor and subcortical networks showed a positive association ($r = 0.201$, $p < 0.05$). Stronger connectivity between dorsal attention and default mode networks indirectly reduced NSSI by enhancing psychological resilience, highlighting resilience as a critical protective factor.

Conclusion: These findings underscore the importance of targeting specific brain connectivity patterns and enhancing psychological resilience as crucial components of neurobiologically informed interventions.

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Introduction

Non suicidal self injury (NSSI) is increasingly viewed not only as a behavioural issue but also with neurobiological underpinnings. Disruptions in brain network connectivity, alongside deficits in psychological resilience, may critically contribute to the development and maintenance of NSSI behaviours (Lloyd-Richardson et al. 2007; Ghinea et al. 2020; Zetterqvist et al. 2020). Investigating these mechanisms could enhance prevention strategies; therefore, this study aims to explore how psychological resilience interacts with resting state brain connectivity to influence NSSI. By adopting a whole-brain approach, we seek to provide a comprehensive understanding of the functional neural networks associated with NSSI and how resilience may mediate these relationships.

NSSI refers to intentional, repetitive self-destructive behaviour without suicidal intent (Lloyd-Richardson et al. 2007). Typical forms include hitting, pinching, cutting, burning, and scratching. In the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), NSSI is recognised as a distinct condition, independent of other mental disorder classifications (Ghinea et al. 2020; Zetterqvist et al. 2020). It is highly prevalent among adolescents and represents a major paediatric mental health concern. The onset and peak prevalence of NSSI are observed around 14–15 years of age (Esposito et al. 2023). In non-clinical samples, the average prevalence of NSSI is ~17.2% among adolescents and 13.4% among young adults, though these rates vary across studies (Lloyd-Richardson et al. 2007; Gu et al. 2020). In China, estimates of NSSI prevalence ranged from 15.0 to 41.5% between 2016 and 2021

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(Dickstein et al. 2015; Gu et al. 2020; Zhao et al. 2023), with a notable increase during and after the COVID-19 pandemic. NSSI is considered a specific predictor of suicide risk and a general predictor of broader psychopathological changes (Ghinea et al. 2020).

Resilience is defined as a protective factor, an individual's cognitive capacity and internal strength to overcome adversity, supported by a persistent mindset, strong self-awareness, and a coherent sense of identity (Luthar et al. 2000; Leys et al. 2021). Resilience evolves as a dynamic cognitive process across the lifespan. In adolescents, it is associated with the ability to navigate adversity effectively and may mediate the relationship between childhood maltreatment and suicidal ideation (Fang et al. 2022). Zhang et al. (2023) reported that higher resilience levels are linked to better outcomes in psychological conditions, such as depression, anxiety, and NSSI (Zhang et al. 2023).

Resting state functional magnetic resonance imaging (Rs-fMRI) is a key tool for investigating intrinsic functional brain networks in clinical research, offering robust patterns with strong test-retest reliability (Li et al. 2024). While task-based fMRI captures state-dependent activation, its findings regarding NSSI vary due to the heterogeneous nature of the paradigms (Dahlgren et al. 2018; Hooley et al. 2020). Neuroimaging studies have identified both structural and functional abnormalities in regions, such as the frontal, temporal, parietal, and cortical areas among individuals with NSSI (Jollant et al. 2018; Kamimura et al. 2022). Huang et al. (2020) reported hyperactivation in the right amygdala, left hippocampus, and left posterior cingulate cortex in those exhibiting NSSI behaviours (Huang et al. 2020). Other findings highlight reduced frontolimbic connectivity and suppressed striatal activity as neural correlates of NSSI (Brañas et al. 2021). Although adolescent-specific studies are limited, they have shown reduced resting state functional connectivity (RSFC) between the default mode network (DMN) and the ventral attention network (VAN), along with disruptions in the frontolimbic system that are key areas linked to emotional dysregulation, a known contributor to NSSI (Auerbach et al. 2021; Lu et al. 2024). Recent studies have also emphasised altered RSFC between the limbic network (LN) and major neurocognitive networks (DMN, control, and attention networks) in individuals with NSSI (Li et al. 2024). While much of the current literature has focused on specific regions of interest (ROIs), such as the amygdala and posterior cingulate cortex (Jollant et al. 2018; Kamimura et al. 2022). A broader whole-brain network approach is necessary to fully understand the neural mechanisms underlying NSSI. Such a comprehensive perspective can help uncover functional connectivity patterns that may not be evident in ROI based analyses.

Few studies have explored the neural correlates of resilience. Eaton et al. (2022) found that increased integration within the DMN was associated with higher resilience (Eaton et al. 2022). Liu et al. identified roles for the DMN and dorsal attention network (DAN) in stress recovery and resilience (Liu et al. 2023). Most recently, Trapp et al. (2023) reported involvement of the right medial prefrontal cortex, right orbitofrontal cortex, DMN nodes, and the right inferolateral temporal cortex in resilience (Trapp et al. 2023). Despite these findings, the specific interactions between brain network dysfunction and psychological resilience in the context of NSSI remain poorly understood. To address this gap, our study adopts a holistic approach by employing whole-brain Rs-fMRI analyses to examine neural connectivity patterns associated with NSSI, alongside assessments of psychological resilience.

Specifically, our objectives are to compare brain connectivity patterns between adolescents with and without NSSI using Rs-fMRI, and to explore how psychological resilience differs between these groups and relates to NSSI behaviour. And examine whether psychological resilience mediates the relationship between key brain connectivity features and NSSI engagement. Our findings will advance the understanding of NSSI and inform the development of evidence-based therapeutic interventions designed to enhance resilience and reduce the risk of NSSI among adolescents. Ultimately, this work seeks to contribute to improved mental health outcomes for vulnerable youth populations.

Methods

Participants

The present study recruited 89 participants (41 males and 48 females) aged 12–18 years with diverse educational backgrounds. Participant recruitment and data collection were conducted at the Department of Depression, Shenzhen Kangning Hospital, Shenzhen, China. This sample size was comparable to that of a previous rs-fMRI study on NSSI and was sufficient for machine learning algorithms (Auerbach et al. 2021; Liu et al. 2023; Li et al. 2024). Shenzhen Kangning Hospital's research ethics committee granted ethical approval (Approval No. 2020-K021-04-1). Informed consent was obtained from participants' parents or caregivers. Figure 1 illustrates the enrolment and data acquisition of participants.

Inclusion criteria

Participants diagnosed with major depressive disorder who had engaged in NSSI within the past year, without any severe mental health conditions (e.g. psychosis,

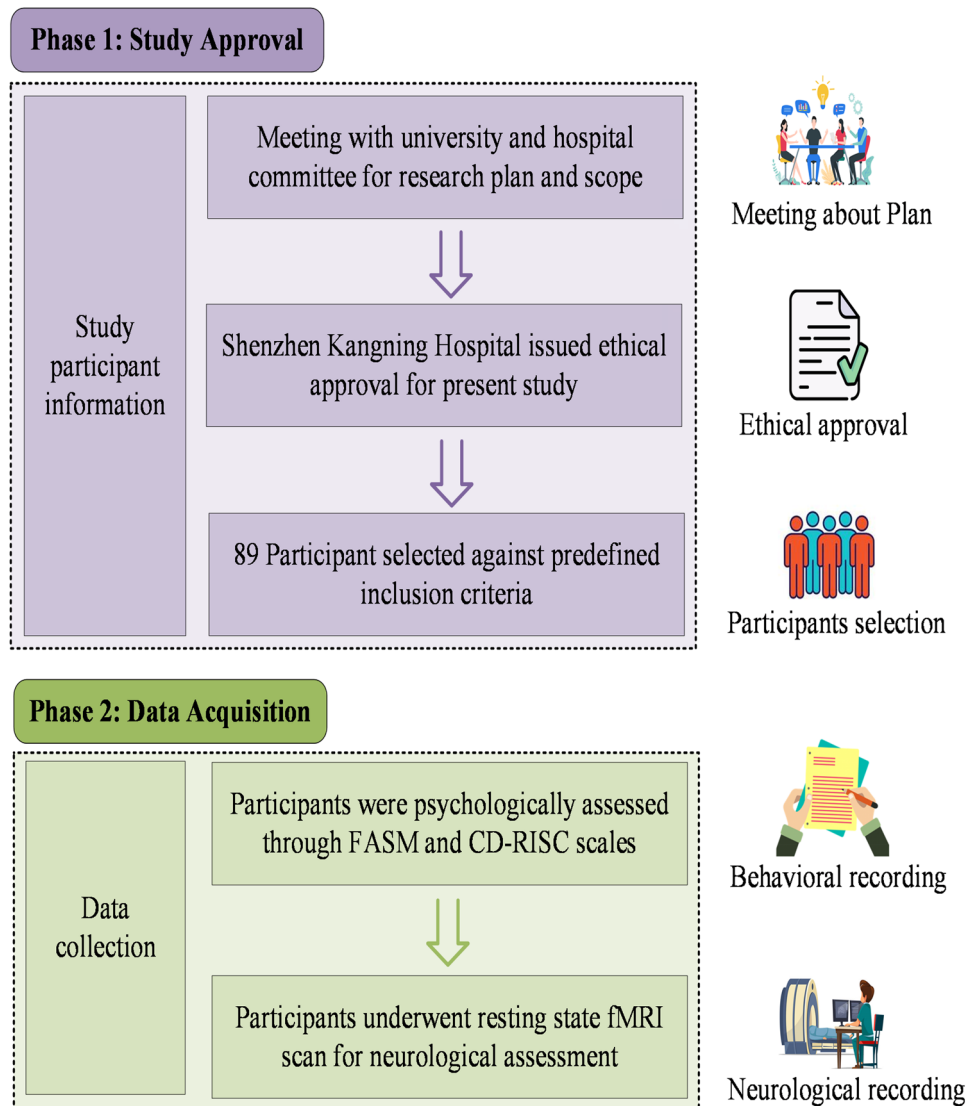


Figure 1. Participants' enrolment and data acquisition process.

bipolar disorder), without substance use, and contraindications to the fMRI scan, were included in the present study.

Exclusion criteria

Participants were excluded if they had no NSSI history within the past year, had any neurological disorders, significant substance use problems, or conditions that made fMRI unsuitable, such as metal implants, severe claustrophobia, or other contraindications.

Data collection

Psychological assessment

Demographic details, including age, gender, and education, were recorded for each participant, along with a psychological assessment. Depression was assessed

using the patient health questionnaire-9 (PHQ-9), functional assessment of self mutilation (FASM), and the Connor-Davidson resilience (CD-RISC) scales—Chinese version were used to assess psychological behaviour as used in previous studies (Ando et al. 2018; Li et al. 2024). A detailed explanation of the psychological scales is provided in the [Supplementary File S1](#).

Neurological assessment (fMRI scan and preprocessing)

For neurological evaluations, fMRI scans were conducted with a 3.0-Tesla MRI scanner (Prisma, Siemens, Germany) at the Department of Radiology, Shenzhen Kangning Hospital. Participants underwent an Rs-fMRI scan (TR/TE = 2000/30 ms; slice thickness = 2.5 mm) and a high-resolution T1-weighted MRI scan (TR/TE = 2530/2.27 ms; slice thickness = 1 mm). The data were preprocessed using the Data Processing Assistant for

Rs-fMRI Toolkit (<http://rfmri.org/DPARF>) in conjunction with Statistical Parametric Mapping (SPM). Additional details regarding data acquisition and preprocessing are described in the [Supplementary File S1](#).

Functional network construction

After preprocessing, to define brain network nodes, the brain was parcellated into 232 cortical and subcortical ROIs. The cortical ROIs ($n=200$) were identified using Schaefer et al.'s scale-200 whole brain functional parcellation (Schaefer et al. 2018), while the subcortical ROIs ($n=32$) were identified based on Tian et al.'s subcortical functional parcellation (Tian et al. 2020). These ROIs were further categorised into eight distinct brain networks: the visual network (VN), somatomotor network (SMN), dorsal attention network (DAN), salience/ventral attention network (VAN), limbic network (LN), frontoparietal/control network (FPN), default mode network (DMN), and subcortical network (SCN) (Yeo et al. 2011). This classification approach is consistent with prior research that utilises large-scale network-based analyses to examine RSFC (Li et al. 2024). The averaged BOLD time series for each brain network was extracted for each subject. Pearson's correlation coefficients were calculated for all pairs of the eight networks, resulting in an 8×8 functional connectivity

matrix for each participant. These correlation matrices were subjected to Fisher's r -to- z transformation. The upper triangular portion of the RSFC matrix, including the diagonal (36 brain features), was selected to make one feature vector for each subject, as illustrated in Figure 2.

Statistical analysis

Descriptive statistics were used to summarise and report the basic characteristics of the data. Point biserial correlation (binary-continuous relationships) was conducted to analyse the relationships between NSSI, psychological resilience, and brain features. An independent two-sample t -test was conducted to compare brain features between the NSSI groups. Binary logistic regression with 5-fold cross-validation was performed to detect risk factors for NSSI and to assess the accuracy of the regression model. SVM with 5-fold cross-validation, in which 80% of the data was used as the training set and the remaining 20% was used as the testing set, was performed to select brain features that show high accuracy with NSSI. A mediation analysis was performed to examine the role of psychological resilience as a mediator between brain features and NSSI. False discovery rate (FDR) correction was applied to adjust p -values derived from multiple

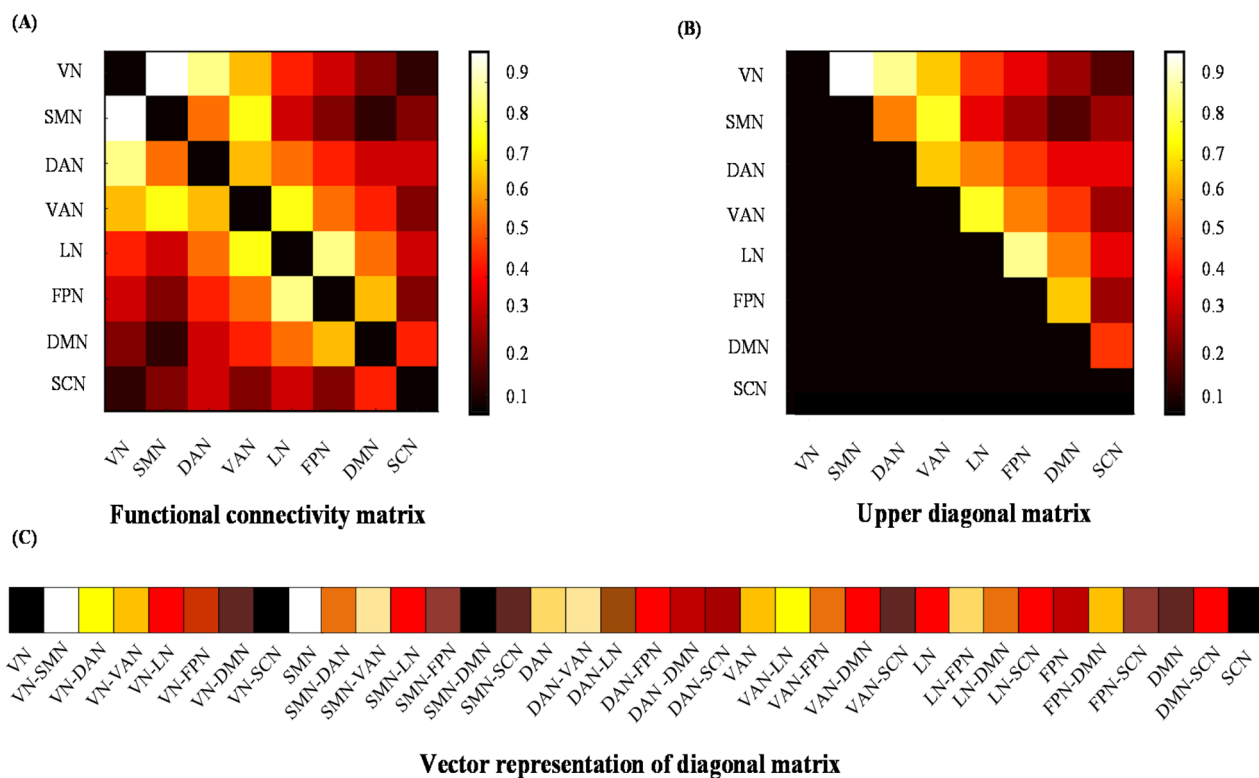


Figure 2. Pipeline of 36 brain features (RSFCs) extraction from fMRI recorded data. (A) Functional connectivity matrix of 8×8 brain networks. (B) Upper diagonal matrix of connectivity matrix. (C) Vector representation of upper diagonal matrix resulted in 36 brain features.

comparisons (Genovese et al. 2002), and $p < 0.05$ was regarded as statistically significant. SPSS (26.0) and MATLAB (R2023a) were used for the analysis.

Results

Reporting of basic characteristics of data

The basic characteristics of the collected data are summarised and presented in Table 1. Overall, participants were 15.39 ± 1.77 years old, with an average education level of 9.75 ± 2.05 years. No significant differences were observed between groups in terms of age, education, and depression scores. In contrast, the distribution of sex differed significantly between groups, with a substantially higher proportion of females being positive for NSSI. Moreover, resilience scores were significantly lower in the NSSI positive group (23.41 ± 7.95) than in the NSSI negative group (27.23 ± 6.52), indicating that lower resilience is associated with the presence of NSSI.

Assessment of strength and direction of the association between brain features, connectivity, and NSSI

After controlling for sex, age, education, and depression (covariates) and applying the FDR correction for multiple comparisons, two significant associations emerged. Point biserial correlation analysis revealed a significant negative association of SMN-LN with NSSI ($r = -0.222^*$, $p < 0.05$) and a strong positive correlation of SMN-SCN with NSSI ($r = 0.201^*$, $p < 0.05$), as shown in Table 2.

Comparison of brain features connectivity between NSSI-positive and NSSI-negative groups

After adjusting for covariates, SMN-LN exhibited significant differences between the two groups ($t = -2.369$,

$p_{\text{FDR}} = 0.020$) as illustrated in Table S1 (Supplementary File). The NSSI-positive group had lower connectivity (-0.0196 ± 0.0278) compared to the NSSI-negative group (-0.0065 ± 0.0244), with a confidence interval (CI) ranging from -0.0284 to -0.0022 , indicating a precise estimate of the mean difference. SMN-SCN also exhibited a significant difference ($t = 2.129$, $p_{\text{FDR}} = 0.036$), with higher connectivity in the NSSI positive group (0.1322 ± 0.0572) compared to the NSSI negative group (0.1090 ± 0.0448), with CI ranging from 0.0068 to 0.0531 .

Determining vulnerable brain features for NSSI and support vector machine (SVM)

Binary logistic regression, after controlling for covariates, evaluated the impact of brain features on NSSI, as presented in Table 3. VN-SCN demonstrated a significant positive association ($\beta = 232.945$, $p_{\text{FDR}} = 0.019$). SMN-VAN was also associated with NSSI ($\beta = 89.475$, $p_{\text{FDR}} = 0.026$); similarly, VAN-DMN was also positively associated ($\beta = 154.889$, $p_{\text{FDR}} = 0.022$). SMN-LN exhibited a significant negative association ($\beta = -109.517$, $p_{\text{FDR}} = 0.006$). DAN also appeared as a significant negative predictor of NSSI ($\beta = -0.154297$, $p_{\text{FDR}} = 0.021$).

Table 2. Strength and direction of the association between NSSI and brain features connectivity.

Brain features	<i>r</i>
VN	-0.043
VN-SMN	-0.128
VN-DAN	-0.149
VN-VAN	-0.077
VN-LN	-0.011
VN-FPN	-0.201
VN-DMN	-0.087
VN-SCN	-0.075
SMN	-0.014
SMN-DAN	0.035
SMN-VAN	0.023
SMN-LN	-0.222*
SMN-FPN	0.005
SMN-DMN	0.144
SMN-SCN	0.201*
DAN	-0.126
DAN-VAN	-0.033
DAN-LN	-0.003
DAN-FPN	-0.002
DAN-DMN	-0.047
DAN-SCN	-0.012
VAN	-0.192
VAN-LN	0.062
VAN-FPN	-0.185
VAN-DMN	-0.016
VAN-SCN	0.003
LN	0.111
LN-FPN	0.131
LN-DMN	-0.010
LN-SCN	-0.115
FPN	0.055
FPN-DMN	0.095

Note: * $p < 0.05$ (2-tailed).

Table 1. Demographic and behavioural characteristics of the NSSI-positive and NSSI-negative groups.

Variable	NSSI positive (<i>n</i> = 46) Mean (<i>SD</i>)	NSSI negative (<i>n</i> = 43) Mean (<i>SD</i>)	<i>t</i> / <i>X</i> ²	<i>p</i> -Value
Age	15.13 (1.86)	15.67 (1.66)	-1.46	0.148 ^a
Education	9.39 (2.26)	10.12 (1.78)	-1.69	0.094 ^a
Sex (male)	19 (41.30%)	22 (51.16%)	15.22	0.001 ^b
Sex (female)	27 (58.70%)	21 (48.84%)		
Depression (PHQ scores)	19.24 ± 4.42	18.64 ± 4.88	-1.43	0.061
Resilience	23.41 (7.95)	27.23 (6.52)	-2.47	0.016 ^a

Note: ^aThe independent samples *t*-test was used to compare the means of two groups for continuous variables.

^bChi-square test was used to determine whether two categorical variables are related to one another.

* $p < .05$.

Table 3. Predicting the brain features' connectivity impact on NSSI through binary logistic regression.

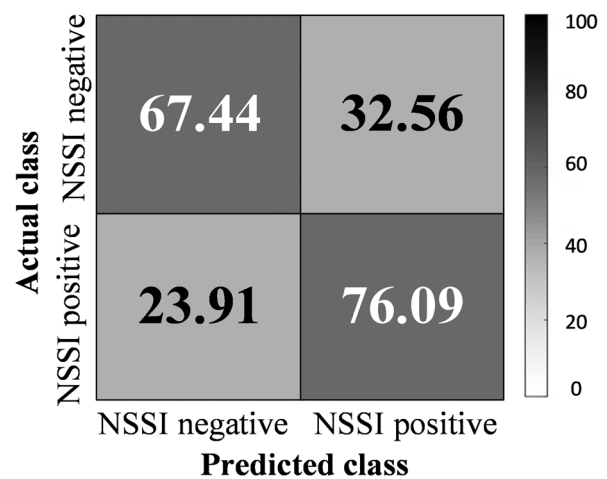
Brain features	β	S.E.	Wald	p_{FDR}
VN	-30.221	23.169	1.701	0.192
VN-SMN	16.745	51.437	0.106	0.745
VN-DAN	-53.502	42.874	1.557	0.212
VN-VAN	-2.79	30.766	0.008	0.928
VN-LN	-63.168	76.028	0.690	0.406
VN-FPN	-31.139	24.817	1.574	0.210
VN-DMN	-113.772	58.319	3.806	0.051
VN-SCN	232.945	99.177	5.517	0.019*
SMN	26.552	40.285	0.434	0.510
SMN-DAN	-30.546	30.993	0.971	0.324
SMN-VAN	89.475	40.144	4.968	0.026*
SMN-LN	-109.517	39.727	7.599	0.006*
SMN-FPN	-18.237	24.086	0.573	0.449
SMN-DMN	103.161	50.007	4.256	0.139
SMN-SCN	2.054	16.281	0.016	0.900
DAN	-154.297	67.094	5.289	0.021*
DAN-VAN	68.908	69.347	0.987	0.320
DAN-LN	-8.995	44.657	0.041	0.840
DAN-FPN	-105.555	61.267	2.968	0.085
DAN-DMN	46.619	31.380	2.207	0.137
DAN-SCN	-43.900	31.830	1.902	0.168
VAN	-14.082	52.485	0.072	0.788
VAN-LM	-89.783	77.954	1.327	0.249
VAN-FPN	-81.618	67.515	1.461	0.031*
VAN-DMN	154.889	95.485	2.631	0.022*
VAN-SCN	-52.812	39.926	1.750	0.036*
LN	20.914	64.790	0.104	0.747
LN-FPN	16.036	34.363	0.218	0.641
LN-DMN	29.032	21.422	1.837	0.175
LN-SCN	-53.874	37.391	2.076	0.150
FPN	22.414	26.625	0.709	0.400
FPN-DMN	30.667	29.164	1.106	0.293
FPN-SCN	-15.921	17.586	0.820	0.365
DMN	23.902	28.979	0.680	0.409
DMN-SCN	-69.321	38.378	3.263	0.031*
SCN	5.405	8.091	0.446	0.504

Note: * $p < 0.05$.

VAN-FPN, VAN-SCN, and DMN-SCN also demonstrated a significant negative prediction value for NSSI with $p > 0.05$. The SVM classifier, trained on connectivity features and covariates, achieved an accuracy of 71.9%, demonstrating robustness to demographic confounds and good predictive performance, as shown in Figure 3.

Mediating role of psychological resilience

Statistical tests evaluate the population-level significance of isolated features, while machine learning models prioritise predictive power based on feature combinations. Our results reflect this dichotomy, with DAN-DMN connectivity not being statistically different between groups, yet contributing most to SVM-based individual-level prediction (Table S2, Supplementary File). Given its prominence, DAN-DMN connectivity was selected for mediation analysis to explore whether psychological resilience mediates its relationship with NSSI. Using a bootstrap method (5000 resamples; $\alpha = .05$), the indirect effect was significant ($\beta = -10.90$, $SE = 6.1350$; 95% CI [-26.0845, -2.0485]), as reported in Table S3, Supplementary File, indicating that greater



Accuracy = $TP + TN / TP + TN + FP + FN = 71.91\%$

Precision = $TP / TP + FP = 72\%$

Recall = $TP / TP + FN = 76.6\%$

F1 score = $2 \times (Precision + Recall / Precision \times Recall) = 74.2\%$

Figure 3. Confusion matrix for the accuracy of the regression model in predicting NSSI.

Accuracy = $TP + TN / TP + TN + FP + FN = 71.91\%$.

Precision = $TP / TP + FP = 72\%$.

Recall = $TP / TP + FN = 76.6\%$.

F1 score = $2 \times (Precision + Recall / Precision \times Recall) = 74.2\%$.

DAN-DMN connectivity associated with higher resilience, which in turn is associated with lower NSSI engagement. The direct path from DAN-DMN to NSSI was not significant ($\beta = 7.01$, $SE = 10.8457$, $p = .517$), supporting the complete mediation. Figure 4 illustrates these direct and indirect mediation pathways.

Discussion

Our research elucidated the complex relationships between the RSFCs of 36 brain features, psychological resilience, and NSSI in adolescents. By integrating neuroimaging data with psychological assessments, we aimed to provide a comprehensive understanding of NSSI, particularly in the context of Chinese adolescents. The findings offer worthy insights into the neurobiological and psychological underpinnings of NSSI, with potential implications for targeted therapeutic interventions.

It is essential to interpret our findings within the context of the study's retrospective design. While associations were found between RSFC, psychological resilience, and NSSI, causality cannot be inferred from these findings. The brain scans were conducted after NSSI behaviours had occurred, and the exact timing between the last NSSI episode and neuroimaging was not uniformly recorded. Consequently, we cannot determine whether the observed RSFC patterns represent predisposing neural features or outcomes of prior self-injurious behaviour.

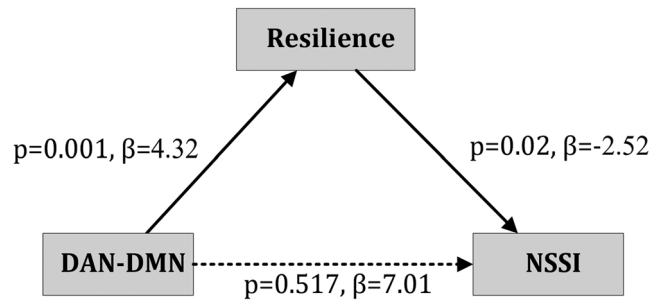


Figure 4. Meditation path of direct and indirect effect of DAN-DMN on NSSI.

The present study found that NSSI is more prevalent in females than in males. Various previous studies have also reported that females are highly vulnerable to NSSI (Ando et al. 2018; Li et al. 2024). The difference in prevalence may stem from how females and males respond to stress. This inequality may be due to sample bias, as the female proportion is higher in our dataset, or gender discrimination. Regarding resilience, the NSSI negative group has a high resilience score. Similar findings were reported by Zhang et al. (2023), who identified resilience as a protective factor against NSSI (Zhang et al. 2023). Another study by Xiao et al. (2020) also reported high resilience in the negative group, supporting our results (Xiao et al. 2020).

SMN-LN exhibits a significant negative correlation with NSSI. SMN comprises regions, such as the parietal lobe, areas of the supplementary motor cortex, primary somatosensory cortex, and primary motor cortex (Soto-León et al. 2022). It plays a role in guiding voluntary movements, coordinating motor functions, processing somatosensory information, and integrating sensory experiences with emotional and cognitive processes (Bi et al. 2022), while the LN, comprising the anterior cingulate cortex, hippocampus, amygdala, and prefrontal cortex, is responsible for emotion processing, control, and memory (Zhou et al. 2022). Zhou et al. (2022) reported that the right medial cingulate gyrus (part of the SMN) is negatively correlated with suicidal ideation (Westlund 2019). Similar to our result, Schreiner et al. (2017) narrated that disrupted connectivity between the amygdala and SMN is associated with NSSI (Schreiner et al. 2017). Otto et al. (2023) also reported alterations in the RSFC of brain areas linked with somatosensory alertness, effortful emotion regulation, and attention among adolescents with NSSI (Cacciola et al. 2017). The diminished RSFC between SMN and LN may indicate impaired integration between emotional processing (LN) and motor planning (SMN), thereby disrupting adaptive somatic responses to distress. This disconnection could paradoxically increase the risk of NSSI as a compensatory

mechanism to re-establish somatic-emotional awareness through physical stimulation.

On the other hand, the SMN-SCN exhibits a significantly high positive correlation with NSSI. SCN encompasses the brainstem areas, thalamus, and basal ganglia, and is responsible for reward processing, motor control, motivation, and emotions (Won et al. 2023). Recently, Won et al. (2023) have demonstrated that self-harming behaviours and other dysfunctional behavioural patterns can be linked to enhanced connectivity within the SCN (Schreiner et al. 2017). An epoch study by Li et al. (2024) reported that NSSI is mainly linked to the disturbed SCN's basal ganglia connectivity (Li et al. 2024). A stronger SMN-SCN connection is positively correlated with increased motor responses to emotional distress, which can lead to the development of NSSI as a coping strategy, as self-injury provides a momentary sense of relief or reward, prolonging the cycle. These findings suggest that altered connectivity in networks implicated in stress response and emotional regulation may be associated with NSSI.

In determining the predictor factor, we concluded that increased connectivity of VN-SCN contributes to NSSI. The VN processes visual information and plays a role in NSSI (Mao et al. 2020). Similar findings were reported in a previous study, which observed alterations primarily in the LN, VN, and SCN in the NSSI-positive group (Brañas et al. 2021). Our findings align with those of Schreiner et al. (2017), who reported that the amygdala, supplemental motor area, and visual attention regions play a role in NSSI occurrence (Schreiner et al. 2017). Vision is an integral part of selective attention; therefore, increased VN-SCN connectivity enhances visual processing and reinforces motor responses to emotional stimuli, causing obsessive behaviours, such as NSSI.

SMN-VAN is a positive and significant predictor of NSSI. VAN has a role in attentional regulation, emotion processing, and orienting to salient stimuli (Han et al. 2023). Han et al. (2023) suggested that attentional and motor networks play a role in self-injurious behaviour,

which supports our findings (Han et al. 2023). Mao et al. (2020) also discovered a favourable positive association between early life stress and connectivity in the VAN, DAN, SMN, and VN (Mao et al. 2020). The increased connectivity between SMN and VAN indicates that these individuals, who are highly sensitive to external stimuli, have a lower threshold for responding to stress or emotional pain through self-injury.

Additionally, it was found that VAN-DMN connectivity was a significant predictor of NSSI. DMN is engaged in self-referential thought and mind-wandering (Ho et al. 2021). Ho et al. (2021)'s findings, which reported alterations in VAN and DMN connectivity associated with NSSI behaviour among adolescents, align with our results (Ho et al. 2021). Likewise, Li et al. (2024) also observed the role of VAN-DMN connectivity in NSSI (Li et al. 2024). Overall, we can conclude that individuals predisposed to NSSI exhibit higher levels of rumination and self-focused attention, which may be attributed to dysregulated interactions within the VAN-DMN.

DAN functions for goal-directed attentional activities (Drabble et al. 2014) and was also a significant negative predictor of NSSI. This finding is consistent with the results of Drabble et al. (2014) and Mao et al. (2020), who identified goal-directed attention as a mechanism for reducing self-harm behaviours (Drabble et al. 2014; Mao et al. 2020). A more robust DAN connectivity helps individuals maintain their attention on objectives and tasks, which may minimise their tendency to engage in impulsive, self-destructive behaviour.

VAN-FPN connectivity is a preventive measure against NSSI. The FPN, also called the control network, contributes to cognitive control and executive functioning (Dixon et al. 2018). The findings of Koenigsberg et al. (2014), who reported that improved executive function and attentional control correspond to reduced rates of self-injurious behaviours, align with our results (Koenigsberg et al. 2014). The negative relationship between DMN-SCN connectivity and NSSI underscores the notion that increased connectivity between these networks facilitates the interaction between self-referential thoughts and motor control, thereby reducing the risk of NSSI. This finding is consistent with research by Andrews-Hanna et al. (2014), which suggests that effective DMN connectivity facilitates improved control over habitual behaviours associated with self-injury (Andrews-Hanna et al. 2014).

The main finding of this study is that psychological resilience statistically mediates the connection between DAN-DMN and NSSI. We chose resilience as the mediator between brain connectivity and NSSI based on evidence from developmental neuroscience and

resilience theory (Rachmad 2022; Liu et al. 2023; Weedage et al. 2025). RSFC patterns like DAN-DMN coupling reflect stable, trait-like neural structures that support attentional control, self-referential processing, and emotional regulation, core neurocognitive processes foundational to the developing and maintaining resilience over time (Eaton et al. 2022; Trapp et al. 2023). In contrast, resilience is a dynamic psychological ability that influences neural responses to environmental stressors in a state-dependent way, rather than fundamentally changing baseline connectivity patterns during adolescence (Luthar et al. 2000; Liu et al. 2023).

The alternative pathway, where connectivity mediates the link between resilience and NSSI, would require resilience to produce lasting changes in intrinsic neural network architecture before influencing behaviour. However, current empirical evidence, especially in adolescents, does not support this directionality (Weedage et al. 2025). Instead, resilience seems to buffer maladaptive outcomes like NSSI by modulating neural reactivity without significantly reshaping networks (Xiao et al. 2020; Zhang et al. 2023). Furthermore, viewing resilience as the mediator is clinically relevant because resilience is a proximal and modifiable target for intervention, whereas baseline connectivity patterns are relatively stable and less directly changeable (Weedage et al. 2025).

While multiple connectivity patterns correlated with NSSI, only DAN-DMN connectivity showed significant mediation through resilience. Higher DAN-DMN connectivity is associated with high resilience and lower risk of NSSI. DAN-DMN impacts NSSI through the indirect path. Moreover, disruption of DAN-DMN connectivity has been identified in several kinds of mental illnesses, suggesting its potential as an indicator of cognitive impairment. The selection of DAN-DMN connectivity for mediation analysis was informed not only by predictive modelling but also by its functional relevance to cognitive control and self-referential processing. Ho et al. (2021) and Wen et al. (2023) reported that disruptions in the normal functioning of the DAN, particularly its interaction with DMN, were connected with increased risks of NSSI (Ho et al. 2021; Wen et al. 2023). Similarly, Liu et al. (2023) also discussed the role of both DMN and DAN in psychological resilience and stress recovery (Liu et al. 2023).

Our mediation model is grounded in evidence that resilience dynamically modulates neurobehavioral pathways. Unlike NSSI behaviours, which are downstream manifestations of distress, resilience reflects a pre-existing capacity to regulate neural responses to adversity (Trapp et al. 2023). This explains why resilience mediates the effects of connectivity on NSSI,

rather than the reverse, consistent with longitudinal studies of adolescent neurodevelopment (Andrews-Hanna et al. 2014). Taken together, our findings support the hypothesis that brain network connectivity, in conjunction with psychological resilience, contributes to the neurobiological profile of adolescents engaging in NSSI.

Conclusion

The study explored the obscure relationship between brain features, connectivity, and NSSI through psychological resilience in adolescents. DAN-DMN connectivity indirectly influenced NSSI *via* resilience, indicating that adolescents with higher resilience are less likely to engage in such behaviour.

This study lays the groundwork for neuroscience-informed interventions aimed at enhancing resilience in adolescents at risk for NSSI. Potential therapies may include neurofeedback and non-invasive brain stimulation techniques. Therapy programs can be customised by integrating neuroimaging profiles with psychological assessments, especially for individuals with low resilience scores. Additionally, combining cognitive behavioural interventions with real-time neurofeedback may help adolescents develop better emotional control and adaptive coping. These strategies aim to enhance resilience by modulating both brain connectivity and behaviour, thereby supporting personalised therapies that reduce NSSI and improve adolescent mental health.

Limitations

With lots of positive aspects, this study also has some limitations. First, the sample was limited to China, and we only used Rs-fMRI. Incorporating task-based paradigms and dynamic functional connectivity analyses with multinational participants in future work would provide a richer, more ecologically valid understanding of the neural mechanisms underlying these behaviours.

Secondly, while we controlled for age, gender, and other factors in our analysis, we acknowledge that psychological resilience can be influenced by a range of other psychosocial factors that were not measured in this study. Future research should incorporate a broader array of psychosocial and contextual factors to gain a more comprehensive understanding of the mechanisms. Another limitation concerns the absence of precise data on the temporal proximity of NSSI episodes to the fMRI scans. Additionally, all participants reported engaging in NSSI within the past year; variations in the recency and frequency of these behaviours

may influence the observed neural patterns. Future studies should incorporate real-time tracking of self-injury behaviour to contextualise RSFC findings better. We also acknowledge that future longitudinal and experimental work should directly compare both possible mediation pathways, in which brain networks should be taken as predictors and mediators.

Ethical approval

The study protocol received ethical approval from the Institutional Review Board of Shenzhen Kangning Hospital (Approval No. 2020-K021-04-1).

Disclosure statement

No potential conflict of interest was reported by the author(s).

Declaration of generative AI and AI-assisted technologies in the writing

During the preparation of this work, the authors utilised the Grammarly tool to enhance their language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Data availability statement

The data used in this study will be made available upon reasonable request, provided a valid official email from the corresponding author is submitted.

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